

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

2-dim mfd $\underline{L(p,q)}$

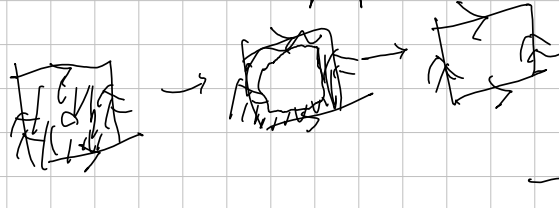
$M \cup N$ $\wedge \vee$ $\langle S, S' \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$

τ_0, τ_1
 $a \xrightarrow{\tau_0} b$
 $\downarrow$
 $\tau_0 \sim \tau_1$
 $\mathbb{Z} \times \mathbb{Z}$
 $\langle a, b \mid aba = b \rangle$
 $ababab \dots$
 $ababab \dots$
 $ababab \dots$

$aba^2b^2 = e$

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

$\langle a, b \mid ab = ba \rangle$
 $\mathbb{Z} \times \mathbb{Z}$



$S^1 \vee S^2$



$\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$

$T: \mathbb{Z} \times \mathbb{Z}$

$T \sim \mathbb{Z} \times \mathbb{Z}$

$\mathbb{Z} \times \mathbb{Z}$

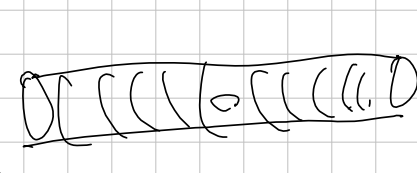
Theorem

$M: n\text{-dim}$
 with $(n \geq 2)$

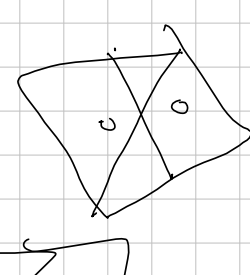
then $\pi_1(M) \cong \pi_1(M \setminus \{x\})$

$\pi_1(RP^n) \cong \mathbb{Z}$
 $(n \geq 2)$

$\pi_1(S^n) = 0$
 $n \geq 2$



$\{Free\}$

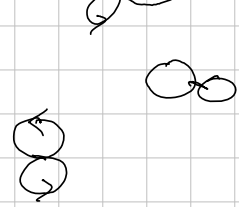
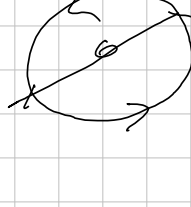


$S^1 \vee S^1$

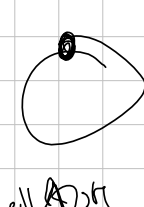
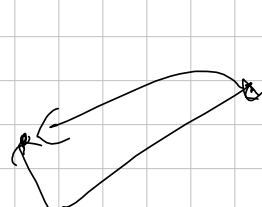
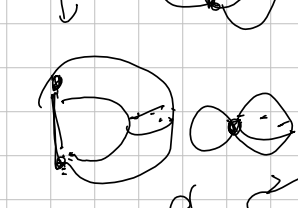
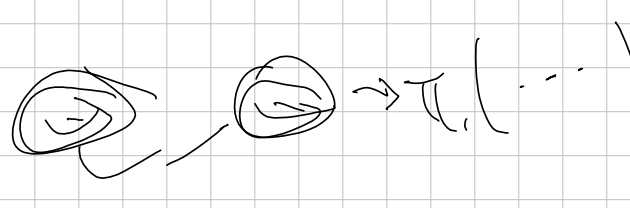
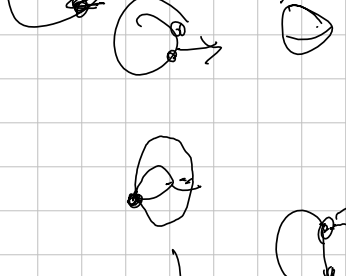
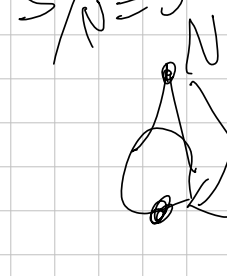
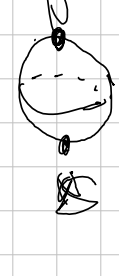
$\mathbb{Z} \times \mathbb{Z}$



$S^1 \vee S^1 \vee S^1$
 $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$

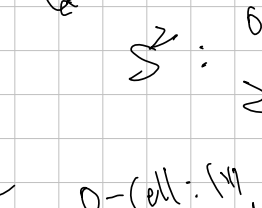


Example



S^1 : 0-cell $\mathbb{Z}$
 $1\text{-cell } \mathbb{Z}$

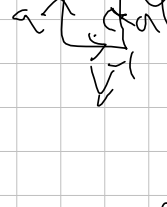
$S^1 \vee S^1$



S^2 : 0-cell $\mathbb{Z}$
 $2\text{-cell } \mathbb{Z}$

$S^1 \vee S^1$

RP^2 : 0-cell: $\mathbb{Z}$
 $1\text{-cell: } \mathbb{Z}$
 $2\text{-cell } \mathbb{Z}$



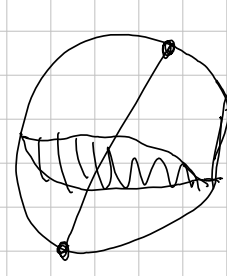
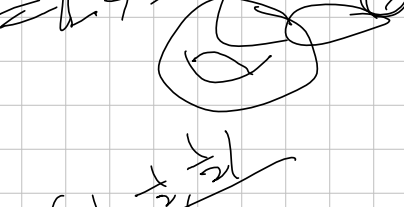
$abab = e$

$RP^2: \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$
 $a^2 = 1$



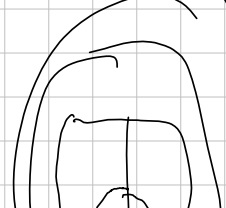
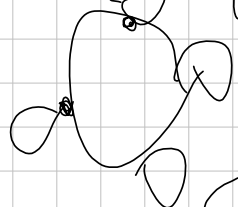
RP^2 : 0-cell $\mathbb{Z}$
 $1\text{-cell } \mathbb{Z}$
 $2\text{-cell } \mathbb{Z}$

$\pi_1(RP^2) = \pi_1(RP^2) = \mathbb{Z}_2$

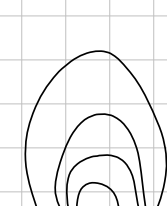
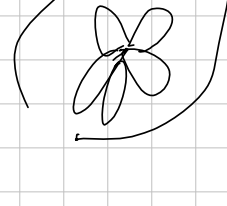
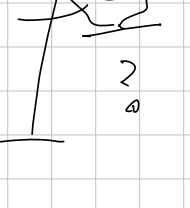
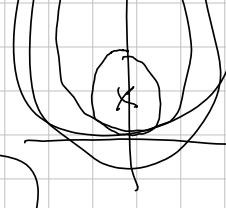
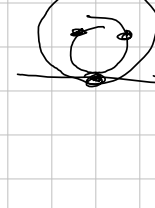


$\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$
 $\cup \mathbb{Z} \times \mathbb{Z}$
 $\cup \{0,0,2\} \sim \mathbb{Z} \times \mathbb{Z}$

$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$



$\mathbb{R}/\mathbb{Z}$



$\mathbb{Z} = \mathbb{Z} \times \mathbb{Z}$



$\mathbb{Z} \times \mathbb{Z}$

$K_1 = \mathbb{Z}$
 $K_2 = \mathbb{Z}$

$2\pi \cdot \mathbb{Z}$

$\int \frac{1}{z} dz$
 $\frac{1}{2\pi i}$